Exploration of enhancer-promoter interactions and their transcription factor networks using topic modelling

Dennis Hecker[1,2,3], Nina Baumgarten[1,2,3] and Marcel H. Schulz[1,2,3]

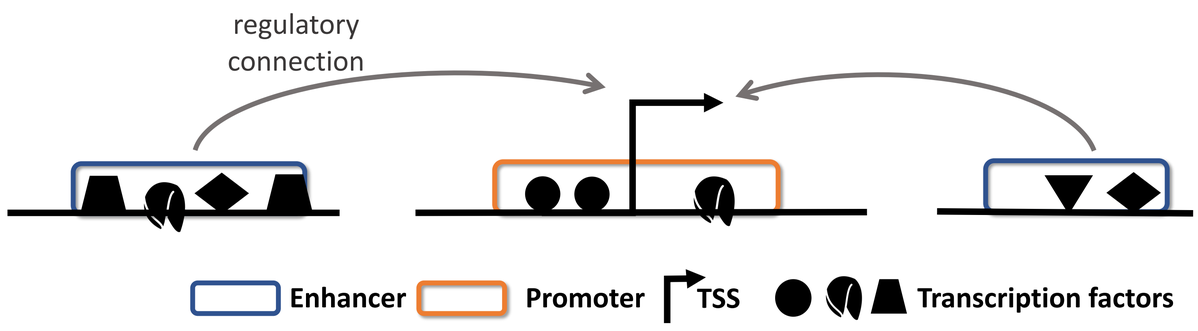
[1]Institute for Cardiovascular Regeneration, Goethe University Hospital, 60590 Frankfurt am Main, Germany, [2]Cardio-Pulmonary Institute, Goethe University Hospital, 60590 Frankfurt am Main, Germany, [3]German Center for Cardiovascular Research, Partner site Rhein-Main, 60590 Frankfurt am Main, Germany

Motivation

Advances in epigenomics have highlighted the importance of non-coding regions in the DNA for regulation of gene expression. Key players in gene regulation are **enhancers**. They can be located far away from their target genes and affect gene expression by serving as **transcription factor (TFs) binding sites**. A gene's promoter is also bound by TFs which form protein complexes together with the associated enhancer-bound TFs to govern gene expression.

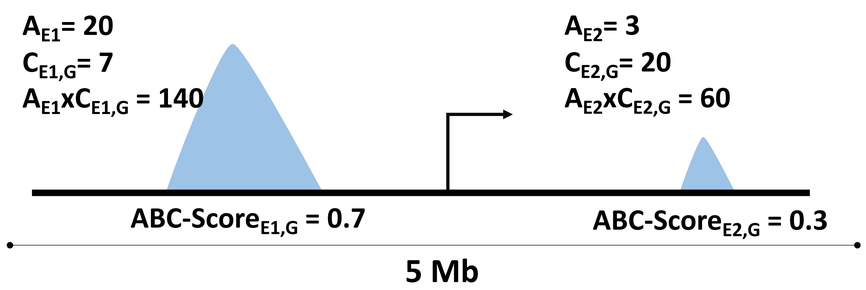
We applied **topic modelling using Latent Dirichlet Allocation (LDA)** on the predicted TF binding sites in enhancers and gene promoters of regulatory connections to explore whether enhancer-promoter interactions can be characterized by their TFs. Although topic modelling has been applied to TF binding before, we follow a different approach by taking both sides of the interaction into account, the enhancer-bound TFs as well as the promoter-bound TFs. This separation could help to describe the formation of regulatory interactions, as we know whether a TF was predicted to bind in the enhancer or the promoter region.

The idea:
Describe enhancer-promoter interactions by their bound TFs



ABC-score to identify promoter-enhancer connections

- Activity-By-Contact model, adapted from Fulco et. al (2019)[1]
- for each gene score enhancer connections in a 5 Mb window around its TSS
- Activity as peak signal to measure strength of regulatory regions, defined by H3K27ac-seq/DNase-seq/ATAC-seq
- Contact as frequency from a Hi-C matrix, Knight-Ruiz normalized
- take the score relative to all enhancers within the window
- score all interactions and consider the top 5% as valid ones

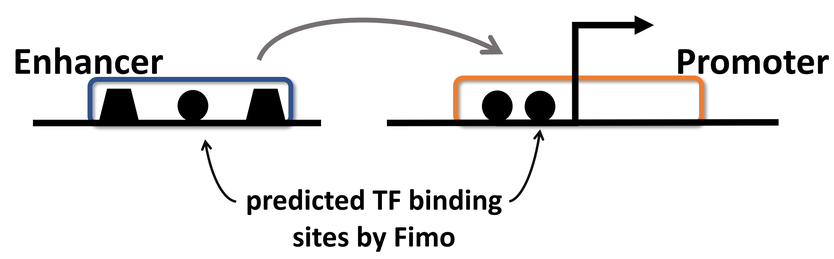


$$\frac{A_E \times C_{E,G}}{\sum_{\text{all e within 5 Mb window}} A_e \times C_{e,G}}$$

Fig. 1 Concept of the ABC-score. All enhancer peaks within a window of 5 Mb around a gene's TSS are scored by their peak signal and their contact frequency to the TSS.

Construct TF count matrix

- create clustered position weight matrices (PWMs) with Mosta[2] to avoid artificial high frequency of similar motifs, resulting in 172 PWMs
- run Fimo[3] on the sequences of enhancer-promoter interactions and count the predicted binding sites
- normalize for sequence length and subtract counts on shuffled sequences



Interactions	Enh - TF ▲	Enh - TF ●	...	Prom - TF ▲	Prom - TF ●	...
Interaction$_i$	2	1		0	1	
⋮						

Topic modelling with LDA

- Latent Dirichlet allocation (LDA) constructs unobserved groups, the so-called topics, to explain similarities in data
- returns two probability distributions:
  - importance of a TF for a topic
  - probability of an interaction to belong to a topic
- can be used in combination with other genomics data to explore the correlation of combinations of TFs with genomic features like gene function, ubiquitousness of genes or histone modifications
- can help in comparing regulatory interactions in between different cell types and conditions and identify relevant TFs
- allows to characterize TFs where no ChIP-seq data is available for

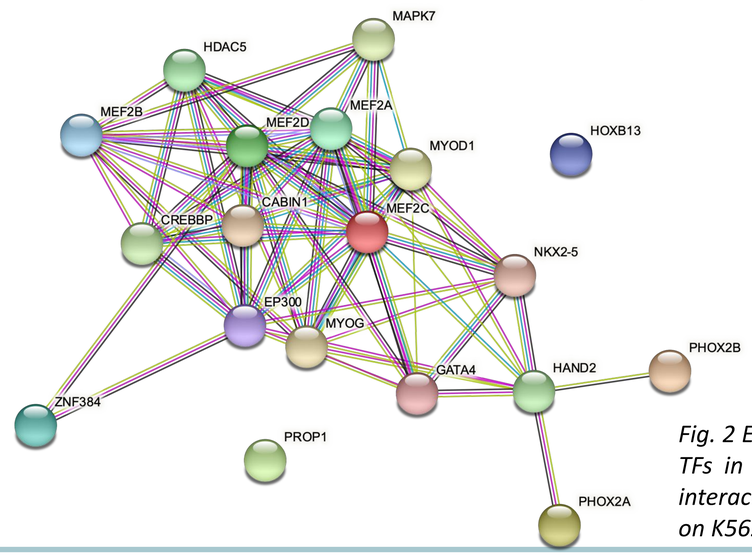


Fig. 2 Example String network of the top 5 ranked TFs in topic 0. String set to a maximum of 10 interactors in the 1st shell. The model was trained on K562 data.

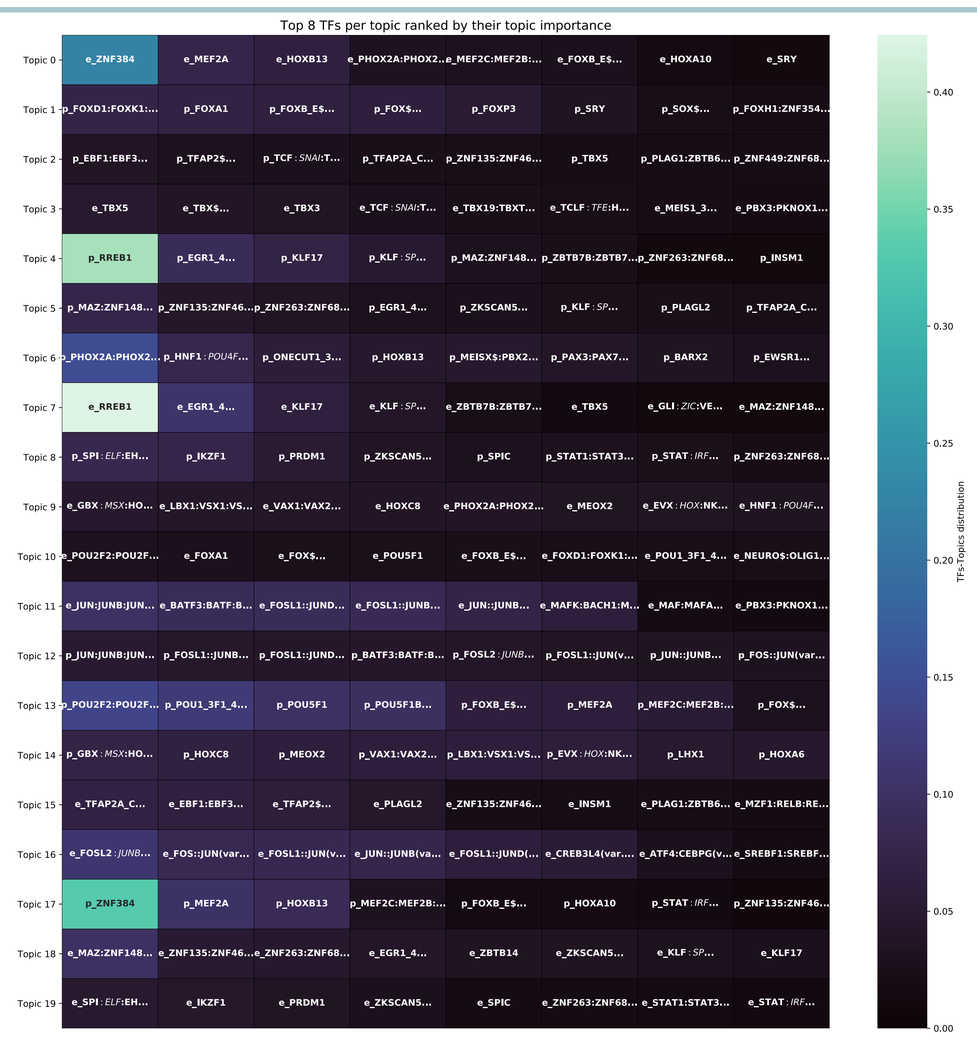


Fig. 3 Top 8 TFs for each topic ranked by their topic importance. 'e_' refers to the TFs binding in the enhancer and 'p_' to the TFs binding in the promoter. The model was trained on K562 data.

Outlook:

- validation with ChIP-seq data where possible
- validation with knock-out experiments
- further dive into the combinations of TFs in the enhancer and the TFs in the promoter
- more extended integration of other genomics data like loops, topologically associated domains, insulator elements etc.

References:

[1] *Fulco, C. P., Nasser, J., Jones, T. R., Munson, G., Bergman, D. T., Subramanian, V., ... Engreitz, J. M. (2019). Activity-by-contact model of enhancer–promoter regulation from thousands of CRISPR perturbations. Nature Genetics, 51(12), 1664–1669. https://doi.org/10.1038/s41588-019-0538-0*

[2] *Pape, U. J., Rahmann, S. & Vingron, M. (2008c) Natural Similarity Measures between Position Frequency Matrices with an Application to Clustering. Bioinformatics, 24 (3), 350–357*

[3] *Grant, C. E., Bailey, T. L., & Noble, W. S. (2011). FIMO: Scanning for occurrences of a given motif. Bioinformatics, 27(7), 1017–1018. https://doi.org/10.1093/bioinformatics/btr064*

Institute of Cardiovascular Regeneration
Goethe-University Frankfurt am Main
CARDIO PULMONARY INSTITUTE
Cardiovascular ncRNA
GOETHE UNIVERSITÄT FRANKFURT AM MAIN
DZHK DEUTSCHES ZENTRUM FÜR HERZ-KREISLAUF-FORSCHUNG E.V.